

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2023

Course: B. Tech. Branch : Mechanical Engineering/Mech(Sandwich)/Production

Subject Code & Name: BTMPE703C, Non -Conventional Machining Semester : VII

Max Marks: 60

Date: 06-01-2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Discuss in detail the various limitations of conventional machining processes.	CO1	6
B) List and explain various factors to be considered for selection of machining process.	CO1	6
C) How the developments in the area of materials are partly responsible for evolution of Advanced Machining Techniques.	CO2	6
Q.2 Solve Any Two of the following.		12
A) Write down limitations and applications of ECM process.	CO1	6
B) Explain the following in Chemical machining process. i)Maskants ii) Etchants	CO2	6
C) Sketch and Explain Electro-chemical Honing.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain need for EBM and mechanism of metal removal in Electron Beam Machining (EBM).	CO4	6
B) What do you understand by Hot Machining? State its applications and limitations.	CO2	6
C) Explain with neat diagram construction and working principle of Plasma Arc Machining.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Define ‘Ultrasonics’ and describe the process in which these are used to machine the material.	CO2	6
B) Describe at least three typical engineering applications of Abrasive Jet Machining.	CO4	6

C)	Explain with neat diagram process parameters in Ultrasonic Machining.	CO2	6
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Q. 5	Solve Any Two of the following.		12
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A)	Discuss about Shaped Tube electrolytic Machining.	CO5	6
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B)	Define 'Hybrid Machining process' and state their applications.	CO5	6
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C)	What is Micro -machining? Discuss its characteristics.	CO6	6
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